Matthew Kuan

BME 212

Self-Designed experiment

The Effect of Chronic Acidic Exposure on Biomechanical Properties of Avian Bone: A Three-point Bending Femoral Failure Analysis

Abstract

This study investigates the direct biomechanical impact of acid exposure on bone integrity with an *ex vivo* model using matched pairs of chicken femora. Each pair was incubated for 20 hours in either distilled white vinegar (pH 2.5) or distilled water as a control (pH 7.0). A three-point bending test was performed to quantify the ultimate force at failure for each bone. Acid-exposed bones exhibited a 15% reduction in force to failure compared to controls ($p = 0.009$), supporting the link between low pH exposure environments and compromised bone strength.

Introduction

The skeletal system plays an essential role in the homeostasis of acidity. Chronic metabolic acidosis, even in mild fluctuations in bodily pH such as acute metabolic acidosis, is known to stimulate bone resorption and inhibit osteoblast activity [1]. This disrupts skeletal integrity and weakens its mechanical properties over time. Dietary patterns can influence the endogenous acid production, so that even individuals without any overt disease could still experience a variation in pH levels that will potentially impact bone metabolism and lead to a weakened skeletal system in what could have been a healthy individual [2,3]. This correlation warrants an investigation into the negative mechanical consequences of acidic environments on bone tissue.

The buffering biomechanisms between acid and the bone's extracellular mineral matrix are insightful into understanding the expected change in mechanical properties. Acid exposure reduces bone carbonate and phosphate content, altering bone composition and decreasing ultimate strength, and increasing fracture risk [4, 5, 6]. Related studies altering the bone properties in different environments offer the opportunity to investigate the pathogenesis of various bone disorders, such as osteoporosis and renal osteodystrophy associated with chronic kidney disease, as well as the possible mitigation of bone loss in certain conditions [7,8]. These conditions are associated with cellular activity regulation, which can be studied through acidosis. We can model this interaction by isolating the physicochemical effects of acidity from more complex cellular and hormonal responses in live organisms [9]. Ultimately, studying the buffering mechanisms of bone to maintain fluctuating pH levels may shed light on the importance of the musculoskeletal system's role in maintaining homeostasis.

We hypothesize that chronic exposure to an acidic environment (≤3 pH) will significantly reduce the force at failure of chicken femora by a minimum of 10% compared to matched controls incubated in a neutral pH environment.

Methods and Materials

Twenty chicken legs were purchased from Stop&Shop. All muscle, tendon, and skin tissue were extracted to isolate the femur. The femora were then assigned by matched pairing based on a similar corresponding bone length into ten different pairs, with the assumption that left and right bones from the same chicken are biomechanically comparable. This design controls for variability between individuals and allows a paired t-test to be used for statistical power in the analysis.

One femur from each pair was assigned to the experimental group and fully submerged in distilled white vinegar (pH 2.50). The contralateral femur from each pair was assigned to the control group and submerged in distilled water (pH 7.00). All bones were incubated in their respective solutions in plastic tupperware at refrigerated temperature (4°C) for 20 hours. After incubation, all bones were removed, organized by pair number in respective groups, patted with paper towels, and allowed to air-dry at room temperature for 2 hours for mechanical testing.

Mechanical properties were assessed using a three-point bending test with an INSTRON 0.025 N (2.5 gf) 5 kN 3433 Tabletop Universal Testing Machine. Bones were placed on two supporting columns with a span length (L) of 40mm. A central loading roller applied a displacement-controlled force at a constant rate of 50 mm/min onto the bone until failure. Force (N) and displacement (mm) were recorded at a high sampling frequency to determine mechanical properties. Assuming hollow cylindrical geometry, the top portion of the bone will undergo compression, and the bottom will be in a state of tension.

From the force-displacement curves, the ultimate force at failure was determined as the maximum force sustained before fracture (F_max). The distributions of F_max values were then compared between the experimental and control groups. Outer and inner radii of the femurs at the breaking point, mid-diaphysis, were measured with digital calipers. A two-tailed paired sample t-test with alpha 0.05 was performed in SPSS to evaluate mean differences in ultimate force at failure between acid exposure to control.

Power Analysis Table

	N[b]	Actual Power[c]	Test Assumptions			
			Power	Std. Dev.[d]	Effect Size	Sig.
Test for Mean Difference[a]	10	.803	.8	.15	1.000	.05

a. Two-sided test.

b. Number of pairs.

c. Based on noncentral t-distribution.

d. Standard deviation of the mean difference.

Table 1. Results of sample power size analysis for two-sided test, matched pairs. Nb = 10 matched pairs (20 bones). Parameters included an assumption of 15% sample difference, 15% standard deviation, 80% power, and alpha = 0.05. N=10 matched pairs calculated.

Matched-pairing balances the variability between groups, and we will assume a parametric distribution. Power analysis to justify sample size was conducted with an estimate of 10% difference in failure force to stay conservative for an affordable sample size and based on a related study assessing acidity on excised murine bones where there was 35% difference at 20% concentration EDTA [10], and an an estimated standard deviation of 15% based on variability for related studies [11], a sample size of 10 matched pairs achieves 80% power at alpha = 0.05.

Results

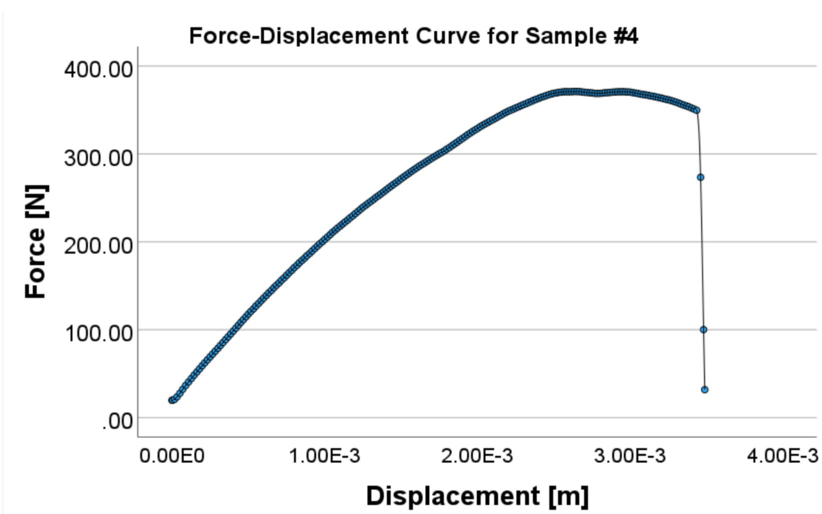

Figure 1. Representative Force-Displacement curve for bone sample #4, tested in neutral aqueous solution. The interpolated curve shows a maximum force of 371.0 N at 0.0026336 m displacement, and will begin plastic behavior.

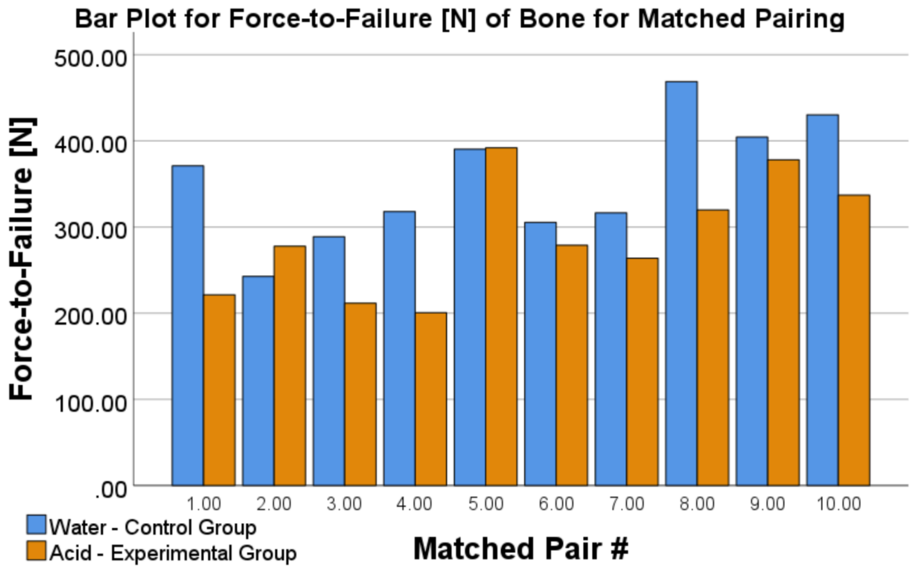

Figure 2. Comparison of matched pairs' force of failure (N). Although matched pairs #2 and #5 showed an increase in strength in the acid group, overall trend favored the control group in mechanical strength in three-point-bending. Mean paired difference = 65.6 N, SD paired difference = 62.8 N. The control group demonstrated a higher mean force at failure (436.2 ± 85.3 N) compared to the experimental group (370.6 ± 78.5 N).

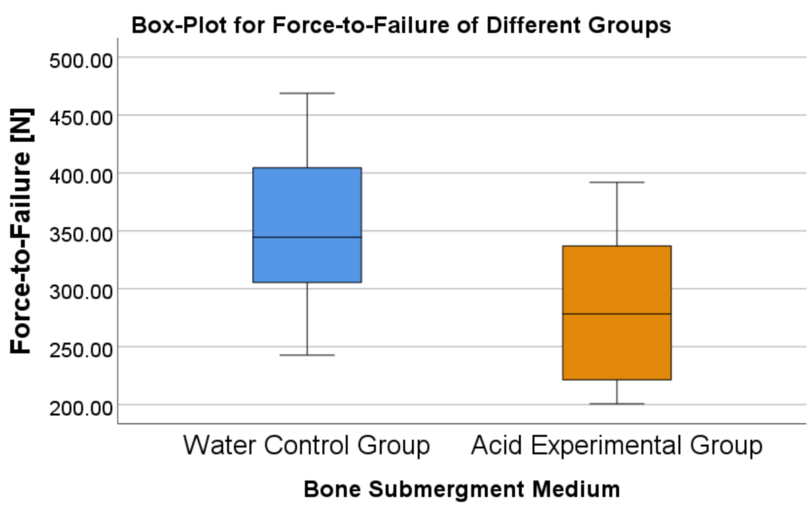

Figure 3. Box plot of force at failure for experimental and control groups. Water group has a median 344.50, variance 4984.672, standard deviation = 70.60, minimum = 242.70, maximum 468.80, interquartile range = 109.65. The acidic group has a median 278.25, variance 4555.07, standard deviation 67.49, minimum = 200.60, maximum = 391.90, interquartile range = 128.30.

The distributions of failure force for both groups were approximately normally distributed, using Shapiro-Wilk = $p > 0.8$ for analysis of pair differences. The mean paired difference in failure force was 65.6 N, with a standard deviation of 62.8N. The two-sided paired t-test indicated statistical significance ($p = 0.0092$).

Discussion

This study demonstrates that chronic exposure to highly acidic environments (pH 2.5) significantly reduces the mechanical strength of avian femoral bones. This is justified by the 15% decrease in ultimate force at failure between both groups, with p-value = $0.0092 < 0.05$. These findings align with our prior research that indicates acidic conditions can demineralize bone [12]. The observed reduction in failure force supports our hypothesis that acid exposure compromises structural integrity. The significant decrease in force-at-failure suggests acid exposure may have reduced the bone's moment of inertia or altered its internal microstructure.

Several limitations should be acknowledged in our experiment. For example, this experimental model with distilled vinegar does not perfectly replicate *in vivo* metabolic acidosis (pH = 5.5-7.0) effects on bone tissue because localized body cells dissolve bone tissue. Additionally, the use of store-bought vinegar introduces other organic compounds, unlike physiological acids; future studies should use controlled pH buffers. Another limitation is that body temperature is not accurately modeled; bone incubation at refrigerated temperature instead of actual physiological temperature may alter results. Other potential limitations include the time delay between drying the bone samples and performing mechanical testing, which may lead to bone brittleness. Finally, while matched pairing controlled for major bone variability, inherent natural variations such as bone length, density, cross-sectional area, and porosity within pairs remain

[13]. Also, the true source of chicken femurs can not be verified unless whole chickens are in the study. However, budgetary constraints and time limited our sample size and access to the full materials that could provide more precise outcomes. Future analysis of interest could include structural stiffness: slope of the linear elastic region, and strain: deformation over length.

Despite these limitations, this experiment provides a clear quantitative understanding of the effect of skeletal system buffering on the mechanical properties of chicken bone. Therefore, there is translational relevance that dietary or metabolic acidosis can weaken bone integrity [14, 15]. Future studies should use more biologically relevant acids (e.g., HCl) at physiological pH ranges and examine results with multiple concentrations of acid levels. Ultimately, there is importance in maintaining systemic acid-base balance for skeletal integrity, and understanding acid environmental effects is a step toward preventing acid-associated bone disorders.

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BME 212 Peer Evaluation Rubric

Your Name: Matthew Kuan

Please rate on a scale of 1-10 with 10 = Excellent and 1 = Poor

Team Member's Name	*Background Research and Development of Project*	*Contribution towards the Proposal*	*Preparation and Execution of Experiment*	*Data Analysis and Interpretation*	**Overall Contribution (Sum of the 4 Columns)**
Matt Chacon	10	10	10	10	40
Isaiah Coombs	10	10	10	10	40
Andrew Muzaka	10	10	10	10	40
Winston Ng	10	10	10	10	40

Comments (optional)

Everyone demonstrated strong collaboration throughout the project, contributing valuable ideas and working effectively together, which enhanced the project's success.